Hierarchical percolation of travel time reveals spatial structure in the Roman road network

Nishanth Kumar[1]

[1][Affiliation – to be completed]

Abstract

We apply hierarchical agglomerative clustering (HAC) on travel-time distances – under single linkage, mathematically equivalent to a percolation sweep over the network's own edges – to the Roman road network, with the aim of recovering spatial units of interest directly from network connectivity rather than from imposed administrative boundaries. Before turning to our own network, we calibrate the method on the exact network and travel-time model used in a recent large-scale reconstruction of the Roman road system, and show that it recovers that network's Roman provinces with an Adjusted Rand Index (ARI) of 0.70 (bootstrap 95% CI [0.68, 0.71]) at a threshold of ∼77 hours (∼3.2 days) of travel – confirming that the method can recover a known administrative partition when one exists in the data. We then apply the same percolation-equivalent clustering to our own, independently constructed network of the Roman road system, where the goal is not to reproduce this calibration result but to characterize the emergent spatial units the network itself suggests. A first, coarse benchmark against the Roman provinces on this network gives a markedly weaker alignment (ARI ≈ 0.43) with a less sharply defined optimal threshold, which we read not as a replication failure but as a signal that provincial administration is only one of several candidate organizing principles the clustering may be picking up – motivating a closer characterization of these units, and a comparison against spatial/temporal null models, as the next step of this work.

1 Introduction

Introduction placeholder – to be completed.

2 Methods

2.1 Data

We work with two networks, referred to throughout by the names of the directories holding their analysis in the accompanying repository: `network1-single` (our own network; Section 3.2) and `network0-average` (the calibration network; Section 3.1). The name suffixes record the HAC linkage setting used for each (see Section 2.2) – this is deliberate, since the two runs are not directly comparable without accounting for it.

network1-single: the itinereX network (primary). Built from the raw Itiner-e linework [de Soto et al., 2025]. Near-miss junctions and duplicate endpoints are snapped, the linework is noded, and degree-2 chains are simplified, giving a weighted graph of 8,460 nodes and 12,849 edges across 12 connected components (see `data_cleaning/` in the accompanying repository). Edge weights are travel times computed from edge length at a constant walking speed of 2 km/h.

network0-average: the calibration network. To check that the method can recover a known administrative partition before applying it to our own network, we separately run the same general approach on the exact Itiner-e export and endpoint-noded topology used by Pažout et al. [2026] (10,506 nodes, 14,770 edges, 13 connected components, including ferry links across the Bosporus and the Dardanelles), with travel time computed from Tobler's hiking function [Tobler, 1993] applied to slope and geodesic edge length. This self-contained calibration run is kept in `results/HAC/network0-average/` and does not otherwise modify the primary analysis.

Province boundaries. Both networks are overlaid with the Ancient World Mapping Center's "Roman Empire AD 200" province polygons, corrected by A. Pažout [Ancient World Mapping Center, 2023] (57 provinces). Each edge is assigned to the province containing its midpoint, falling back to the province with the longest geometric overlap when the midpoint falls outside every polygon; nodes inherit the province of their incident within-province edges. Edges that cross a provincial boundary, or that cannot be assigned to any province, are excluded from the alignment scoring described below.

2.2 Hierarchical agglomerative clustering and percolation

Following the general approach of Arcaute et al. [2016], who use a percolation sweep over distance thresholds between street intersections to recover the hierarchical regional and urban structure of Britain, we apply an equivalent percolation sweep to travel time on the Roman road network itself, rather than to Euclidean distance between unconnected points.

Within each connected component, we compute pairwise travel-time distances between nodes and perform hierarchical agglomerative clustering (HAC; `scipy.cluster.hierarchy.linkage`). For `network1-single` we use *single* linkage: because single linkage merges clusters at the weight of the minimum edge connecting them, cutting the resulting dendrogram at a travel-time threshold t reproduces exactly the connected components of the network with all edges of weight greater than t removed – i.e. HAC under single linkage is mathematically equivalent to a percolation sweep over the network's own edges, computed once per component rather than independently at every threshold. This equivalence is the reason we adopt single linkage for the primary analysis: the resulting clusters are, by construction, genuine percolation clusters of the road network, not an artefact of a particular clustering criterion.

The calibration network, `network0-average`, instead uses *average* linkage, which clusters by mean pairwise travel time between cluster members and does not share the exact percolation equivalence described above. This difference predates the present analysis and is retained here deliberately, flagged by the `-average`/`-single` suffixes on each network's directory name, so that ARI/NMI values reported for the two networks in Section 3 are never read as directly comparable without that caveat in mind.

Sweeping t across the full range of merge heights gives, at each threshold: the number of clusters, and the size of the largest cluster relative to the network (the percolation curve). We additionally fit a power law to the percolation curve over its emergent region.

Alignment with provinces. At each threshold t, we compare the resulting cluster labels to the province labels of the nodes with a known province assignment, using the Adjusted Rand Index [ARI; Hubert and Arabie, 1985] and Normalized Mutual Information [NMI; Danon et al., 2005]. ARI ranges from 0 (random partition) to 1 (perfect agreement, up to cluster relabelling); NMI ranges from 0 to 1 and does not penalize a finer partition as harshly as ARI. We search exhaustively over every distinct merge height for the threshold that maximizes ARI against the provinces, and estimate a 95% confidence interval on that best ARI (and on the threshold it occurs at) by bootstrap resampling nodes with replacement and re-selecting, for each resample, the best-scoring threshold among those closest to the full-data optimum.

3 Results

3.1 Calibration: recovering a known administrative partition (network0-average)

Before asking what spatial units our own network suggests, we first check that the method is capable of recovering a known partition at all, on a network and travel-time model we did not build ourselves. Figure 1 shows the percolation curve and snapshot cluster maps for the calibration network of Pažout et al. [2026], with Tobler-function travel times and AWMC provinces (10,430 of 10,506 nodes have a known province).

Table 1 reports ARI and NMI at six representative travel-time snapshots. Alignment with the provinces rises from essentially zero at 6–12 hours, peaks around 1–3 days, and falls off again as clusters grow beyond provincial scale.

Table 1: Province alignment at fixed travel-time snapshots, calibration network (network0-average; Pažout et al. [2026], Tobler-function travel time; n = 10,430 nodes with known province).

Snapshot	Clusters	ARI	NMI
6 h	2,809	0.039	0.640
12 h	1,376	0.092	0.683
24 h	514	0.201	0.734
2 d	154	0.516	0.803
7 d	22	0.493	0.778
12 d	15	0.308	0.689

An exhaustive search over every merge height finds a single, sharp peak at $t = 77.1$ hours (∼3.2 days), with ARI = 0.695 (95% bootstrap CI $[0.681, 0.710]$). The peak is stable: every bootstrap resample of the node set selects the same 77-hour merge height as optimal. At this threshold the network's emergent regions correspond closely to the Roman provincial map.

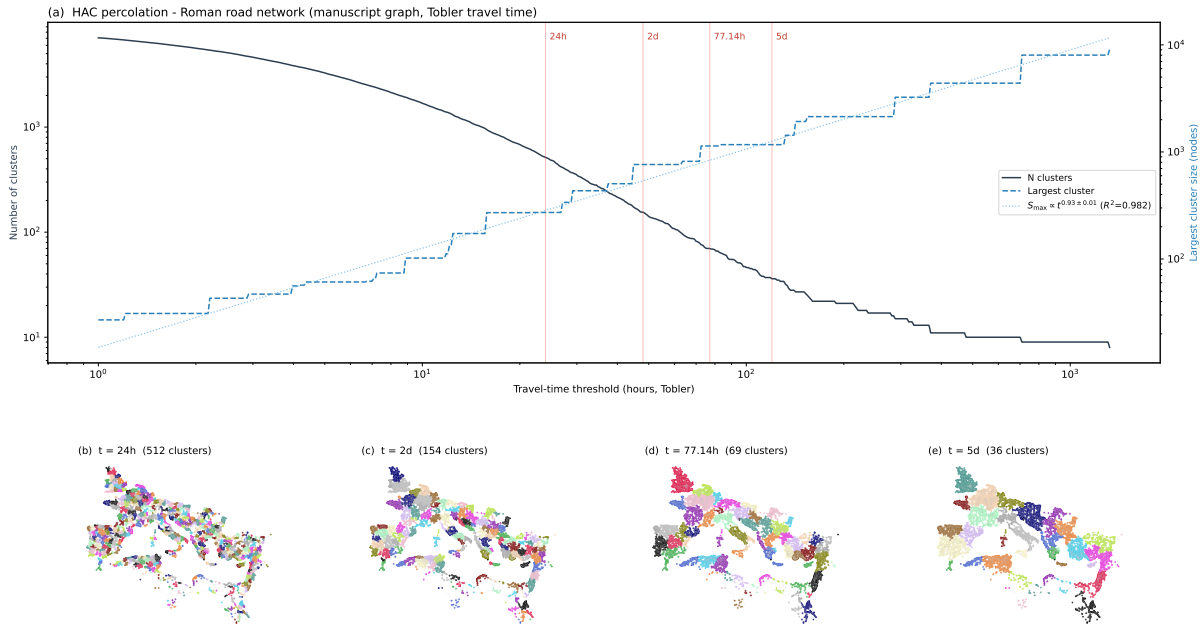


Figure 1: Percolation curve and snapshot cluster maps for the calibration network (`results/HAC/network0-average/maps/hac_percolation_combined.pdf`).

3.2 Towards emergent spatial units in the itinereX network (network1-single)

Having confirmed in Section 3.1 that percolation-equivalent HAC can recover a known administrative partition when one is present, we turn to the question this paper is actually about: what spatial units does the itinereX network itself suggest, under its own independently built topology and a simple constant-speed (2 km/h) travel-time model? This is not a replication of Section 3.1 – we are not trying to reproduce its ARI on a different network, and treating the comparison that way would misread what each run is for. Section 3.1 establishes that the method works; this section applies it, for the first time, to a network nobody has clustered before, with the goal of characterizing the clusters themselves as candidate spatial units, of which provincial alignment is only one possible interpretation among several worth testing.

As a first, coarse orientation, we benchmarked the resulting clusters against the Roman provinces using the same ARI/NMI procedure as Section 3.1: an exhaustive search over every merge height gives a best alignment of ARI $= 0.433$ (NMI $= 0.763$, 291 clusters) at $t \approx 31.8$ hours, with a second, comparably good local optimum around 25 hours rather than the single sharp peak seen in Section 3.1 (95% bootstrap CI on ARI $[0.426, 0.448]$; on the threshold, $[24.8, 31.8]$ hours). We do not read this as a weaker version of Section 3.1's result – on a network built under a different travel-time model, with its own topology, there is no reason to expect the same number – but as a preliminary signal that provincial administration is detectable in, without fully explaining, the clusters this network produces. We have deliberately not built a figure or table around this number here: a single province-alignment statistic does not do justice to what the clusters at each threshold actually look like, and a closer, better-presented look at them is needed before drawing conclusions from it.

What we take this to mean. The more interesting question this result raises is not "why is the ARI lower here", but what the two comparable local optima, and the clusters themselves, actually correspond to – whether that is a different, non-provincial regionalization of the network (e.g. against the Pleiades regions already attached to this network's nodes), a boundary effect of the constant-speed travel-time model, or a genuine feature of the network's own topology. Distinguishing between these is the purpose of comparing this run against the spatial/temporal null models in `results/Null-models/`, which is ongoing work and not yet part of this manuscript.

4 Discussion

Discussion placeholder – to be completed. Note for later: the calibration run (Section 3.1) establishes the method works; the open question for network1-single is what its emergent clusters represent, which needs (i) a proper figure/table once the clusters have had a closer look, and (ii) comparison against the null models in `results/Null-models/` to separate a genuine topological signal from an artefact of the constant-speed travel-time model.

Data accessibility

All code and data underlying this analysis are available in the accompanying repository, under `results/HAC/network1-single/` (primary analysis) and `results/HAC/network0-average/` (calibration analysis).

Authors' contributions

To be completed.

Competing interests

The authors declare no competing interests.

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References

Pau de Soto et al. Itiner-e: a digital gazetteer and dataset of the Roman road network. *Scientific Data*, 2025. https://www.nature.com/articles/s41597-025-06140-z.

Adam Pažout, Tom Brughmans, Pau de Soto, Matteo Mazzamurro, María Coto-Sarmiento, Céline Filet, Eduardo Herrera Malatesta, Mette Louise Nielsen, Gøril Eide Ølgaard, and Peter Birkholm Vahlstrup. Network science reveals the structure and lasting impact of the Roman road system. *Nature Communications*, 2026. In press.

Waldo Tobler. Three presentations on geographical analysis and modeling. Technical Report 93-1, National Center for Geographic Information and Analysis, 1993.

Ancient World Mapping Center. Roman Empire AD 200 provinces, corrected by Adam Pažout. https://github.com/AWMC/geodata/tree/master/Cultural-Data/political_shading/roman_empire_ce_200_provinces, 2023.

Elsa Arcaute, Carlos Molinero, Erez Hatna, Roberto Murcio, Camilo Vargas-Ruiz, A. Paolo Masucci, and Michael Batty. Cities and regions in Britain through hierarchical percolation. *Royal Society Open Science*, 3:150691, 2016. doi: 10.1098/rsos.150691.

Lawrence Hubert and Phipps Arabie. Comparing partitions. *Journal of Classification*, 2(1): 193–218, 1985.

Leon Danon, Albert Díaz-Guilera, Jordi Duch, and Alex Arenas. Comparing community structure identification. *Journal of Statistical Mechanics: Theory and Experiment*, 2005(09): P09008, 2005.